Study Breaks and Perceived Learning

Exploratory Data Analysis

STA304H5F: Surveys, Sampling and Observational Data
Fall 2026

1 Study design and data preparation

This report follows the revised group proposal. Its three planned analyses are:

1. **RQ1: Simple linear regression.** Is break count per session associated with agreement that breaks improve efficiency in the next study cycle?
2. **RQ2: Simple linear regression.** Is break duration linearly associated with perceived next-day retention among break takers?
3. **RQ3: One-way ANOVA.** Does mean perceived next-day retention differ across four mutually exclusive exercise/gaming profiles?

The CSV contains 57 records and 13 columns, dated October 5–9, 2026. Retention is a next-day self-rating (1–10). Break duration is numeric in minutes; no category midpoints are imputed. Efficiency is agreement with a benefit statement (1–7), rather than measured productivity.

Sampling. Recruitment through the course Google Form exchange produced a voluntary-response sample. No random sample of the course was drawn. Self-selection and screening limit generalization to all Fall 2026 STA304H5 students.

Table 1: Preparation rules and analysis denominators.

Issue	Action
Unavailable course IDs	Exclude records 8 and 38, whose IDs are missing. Remaining IDs are not verified against the course roster.
Inconsistent session times	Exclude records 12, 17, 30, 33, 43, 48 and 50: break count times duration meets or exceeds total session duration.
Weekly-hour outliers	Set four impossible values (900, 2,800, 600, 600) to missing in this field only. One unresolved range is already missing.
Undefined break duration	Set duration to missing for both zero-break respondents. One additional duration is missing from an unresolved range; no midpoint is imputed.
Analysis sizes	57 source records; 9 record exclusions; 48 screened records; 46 break takers; 45 complete RQ2 cases; 43 usable weekly-hour values.

The source CSV is unchanged. Timing flags are conservative analysis exclusions, not proof that responses are invalid. Record 25 reports a 780-minute session and 120-minute breaks; it is unusual but internally possible and is retained. Grade values are band proxies, not exact marks. IDs, timestamps and quality notes are administrative fields. Source-record numbers start from 1 after the CSV header; Appendix B visualizes the timing and weekly-hour checks.

2 Summary statistics

Table 2: Numeric summaries of the screened sample.

Measure	n	Missing	Mean	Median	SD	IQR	Min	Max
Study session duration (min)	48	0	150.62	120.00	109.44	60.00	30.00	780.00
Breaks per study session	48	0	2.35	2.00	1.64	2.00	0.00	8.00
Duration of each break (min)	45	3	16.89	10.00	18.81	7.50	5.00	120.00
Next-day retention rating (1-10)	48	0	6.15	6.00	1.40	1.00	3.00	9.00
Weekly study time (hours)	43	5	15.81	15.00	9.61	12.00	5.00	40.00
Prior grade: band proxy (%)	44	4	74.77	75.00	8.49	7.50	55.00	95.00
Break benefit agreement (1-7)	48	0	5.73	6.00	1.16	1.00	2.00	7.00

Generated with `datawizard::describe_distribution()` from the easystats ecosystem. SD is the sample SD; IQR uses type-6 quantiles explicitly. Three unavailable break durations comprise two structurally undefined no-break cases and one unresolved range. Grade proxies and ordinal ratings are summarized numerically without assuming exact interval measurement.

Median session length is 120 minutes, with two breaks per session. Break duration has median 10 minutes, mean 16.89 and SD 18.81, indicating a long upper tail. Retention averages 6.15 (SD 1.40). Valid weekly study time averages 15.81 hours ($n = 43$). Similar means and medians alone do not establish normality.

Table 3: Prior course grade band.

Grade band	n	% of 48
50-59%	1	2.1
60-69%	10	20.8
70-79%	25	52.1
80-89%	5	10.4
90-100%	3	6.2
Missing / unavailable	4	8.3

Table 4: Break benefit agreement.

Agreement	n	% of 48
Strongly disagree	0	0.0
Disagree	2	4.2
Slightly disagree	0	0.0
Neutral	4	8.3
Slightly agree	8	16.7
Agree	23	47.9
Strongly agree	11	22.9
Missing / unavailable	0	0.0

The 70–79% grade band was most common (52.1%). Forty-two respondents (87.5%) at least slightly agree that breaks help efficiency. Four prior grades are unavailable; the recorded data do not establish why. Percentages use all 48 screened respondents, including missing cases; rounding may prevent exact totals of 100%.

Table 5: Recorded year/attention-check item responses.

Recorded response	n	% of 48
Recorded code 1	5	10.4
Recorded code 2	0	0.0
Recorded code 3	29	60.4
Recorded code 4	14	29.2
Recorded code 5	0	0.0
Missing / unavailable	0	0.0

Counts describe recorded responses, not verified years of study. Interpretation as an attention check requires confirmation of the administered prompt.

3 Response distributions and visual exploration

Table 6: Reported break activities.

Activity	n	% of 48
Eating	17	35.4
Exercise	9	18.8
Gaming	17	35.4
Napping	11	22.9
none	1	2.1
Instagram write-in	1	2.1
Scrolling	1	2.1
Scrolling Phone	1	2.1
Social	37	77.1
Space out	1	2.1
Youtube	1	2.1

Table 7: Reported study locations.

Location	n	% of 48
Cafe	4	8.3
Home	41	85.4
Lecture hall	6	12.5
Library	35	72.9
Other campus building	2	4.2
Outdoors	1	2.1
Public transit	3	6.2
Missing / unusable	1	2.1

Multiple selections are counted once per respondent per label; percentages can exceed 100%. All selections are retained, including those exceeding the requested two. "Social" remains the source label without an assumed activity type. The Instagram write-in is shortened for display. Location splitting preserves commas inside parentheses, and campus-building aliases are combined. Activities have no missing responses; one location is unusable.

"Social" is selected by 77.1% of respondents; home (85.4%) and library (72.9%) are the most frequently reported locations. These categories overlap.

Table 8: Distribution of perceived next-day retention.

Rating	1	2	3	4	5	6	7	8	9	10
Count	0	0	3	4	4	17	13	6	1	0
% of 48	0.0	0.0	6.2	8.3	8.3	35.4	27.1	12.5	2.1	0.0

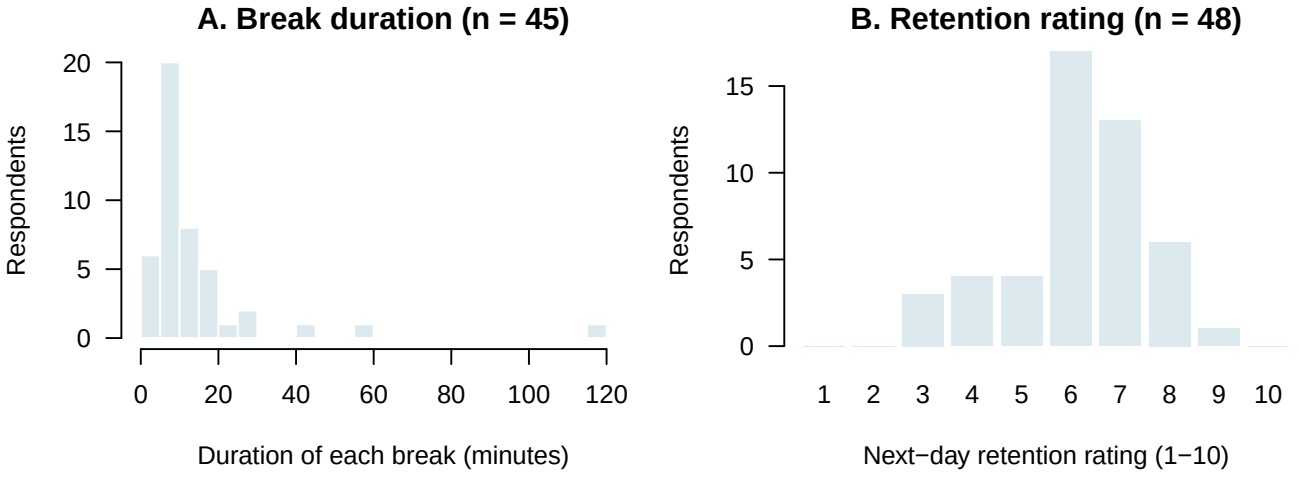


Figure 1: Distributions of break duration and retention. A: observed break duration for the 45 RQ2 cases; the isolated 120-minute value creates a long right tail. B: retention for all 48 screened respondents, with 62.5% rating 6 or 7. Retention is bounded and discrete. Normal-error inference concerns model residuals, not normality of the predictor.

4 Planned tests and RQ1: Simple linear regression

The three proposal analyses use $\alpha = 0.05$. Raw and Holm-adjusted p-values are reported for this family of three tests. Assumption limitations remain relevant.

Table 9: Tests specified in the revised proposal.

Question	Test	n	Statistic	Raw p	Holm p
RQ1	Simple linear regression	48	1.93	0.0594	0.1783
RQ2	Simple linear regression	45	-0.07	0.9445	0.9445
RQ3	One-way ANOVA	46	0.98	0.4128	0.8257

Statistics are slope t with 46 and 43 df, and ANOVA F with (3,42) df.

Model and result. Regress the seven-point benefit-agreement score on the numerical break count per session; test $H_0 : \beta_1 = 0$ against $H_A : \beta_1 \neq 0$. The estimated slope is 0.19377 points per additional break (95% CI: -0.00804, 0.39557), with $R^2 = 0.0751$. The positive fitted association is not significant at 0.05 (raw $p = 0.0594$), before or after adjustment. It does not measure an actual gain in efficiency.

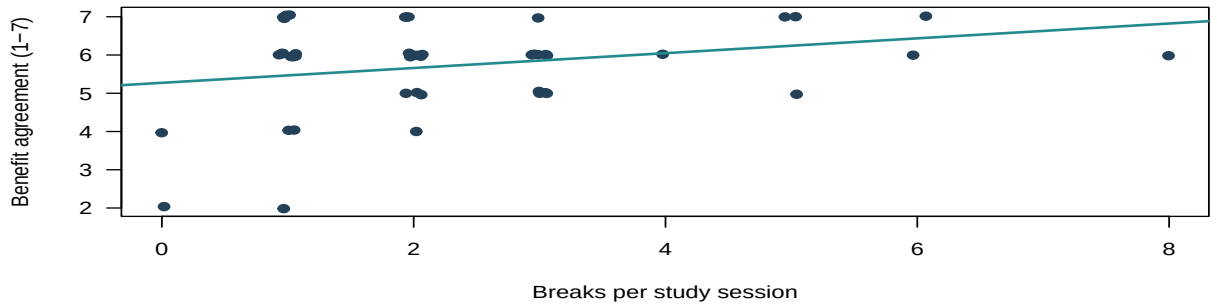


Figure 2: Break count and benefit agreement. Jitter separates overlapping points; the fitted line uses unjittered observations. The fitted direction is positive but uncertainty includes a zero slope.

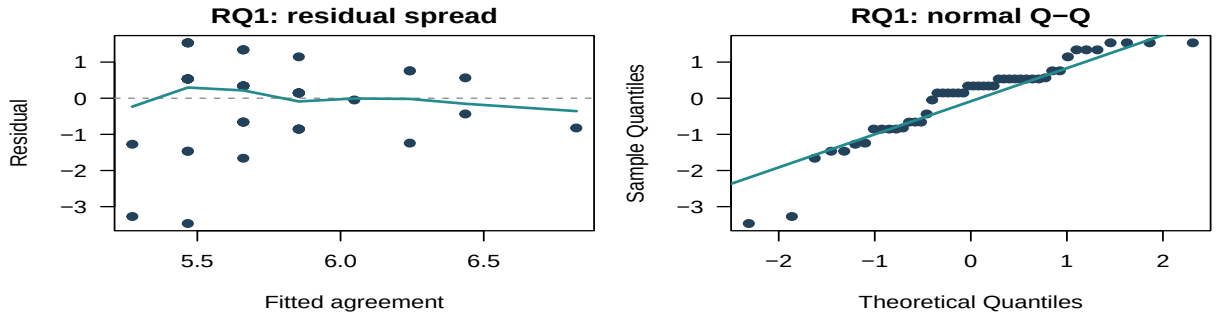


Figure 3: RQ1 residual spread and normal Q–Q plot. Uneven spread and departures from the reference line limit normal, constant-variance inference.

Assumptions. Shapiro–Wilk $p = 0.00085$ and the variance score test $p = 0.00484$ indicate departures from normality and constant variance. The bounded, ordinal response also limits the linear approximation. Independence cannot be established from these plots. Interpret the classical interval and p-value cautiously.

5 RQ2: Simple linear regression

Let X_i be break duration in minutes and Y_i the next-day retention rating. For the 45 complete break takers, fit

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

The primary test is two-sided at $\alpha = 0.05$: $H_0 : \beta_1 = 0$ versus $H_A : \beta_1 \neq 0$. This is the regression slope test planned in the proposal. The model treats the rating scale as approximately numerical.

Table 10: Ordinary least-squares coefficients and 95% confidence intervals.

Term	Estimate	SE	t	p	CI lower	CI upper
Intercept	6.21302	0.27670	22.45421	< 0.00001	5.65501	6.77103
Break duration (min)	-0.00077	0.01101	-0.07000	0.94452	-0.02298	0.02144

$n = 45$; residual df $= 43$; $R^2 = 0.000114$, adjusted $R^2 = -0.02314$; residual standard error $= 1.374$ rating points. Conventional intervals assume the regression error assumptions.

The fitted equation is

$$\widehat{\text{retention}} = 6.21302 + (-0.000771) \times \text{break minutes}.$$

The slope is -0.000771 rating points per minute (95% CI [-0.02298, 0.02144]), $t(43) = -0.070$, $p = 0.9445$. A 10-minute longer break corresponds to an estimated -0.0077-point difference (95% CI [-0.2298, 0.2144]). The fitted direction is slightly negative but statistically indistinguishable from zero; almost none of the variation is explained. This does not establish that no association exists.

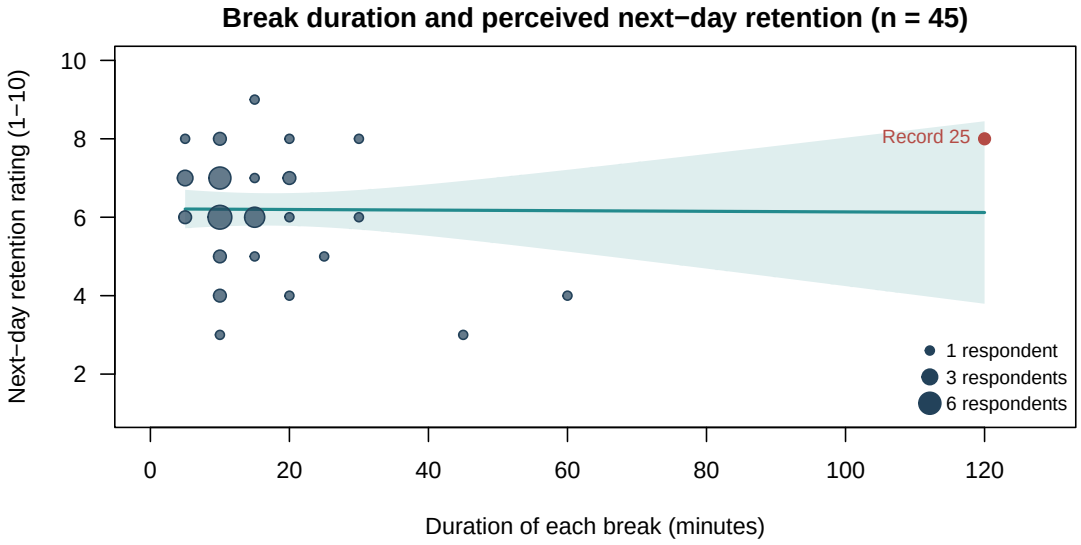


Figure 4: Retention versus duration and fitted regression. Point area shows the number of identical responses. The teal line is the fitted conditional mean; shading is its conventional pointwise 95% confidence band, not an interval for individual responses. The sparse upper-duration range and record 25 make the near-flat line sensitive to influence.

The prominent predicted values at zero minutes, outside the observed 5–120 minute range, are not a comparison with students taking no breaks, who are excluded from this model because their break duration is undefined.

6 RQ3: One-way ANOVA of activity profiles

The revised proposal compares four mutually exclusive exercise/gaming profiles: exercise without gaming, gaming without exercise, neither, and both. This definition handles the multiple-selection question without duplicating respondents across ANOVA groups. Other activities may co-occur in every group, so the comparison concerns these profiles rather than every individual activity type.

All 46 screened break takers enter this comparison, including one with missing break duration. Each contributes once. "Only" below refers to exercise versus gaming, not the absence of all other activities. The ambiguous label "Social" is not recoded or used to assign profiles.

Table 11: Perceived next-day retention by exclusive activity profile.

Profile	n	Mean	SD	Median
Exercise only	5	6.20	0.45	6.00
Gaming only	12	5.75	1.54	6.00
Neither	25	6.32	1.35	7.00
Both	4	7.00	1.41	7.50

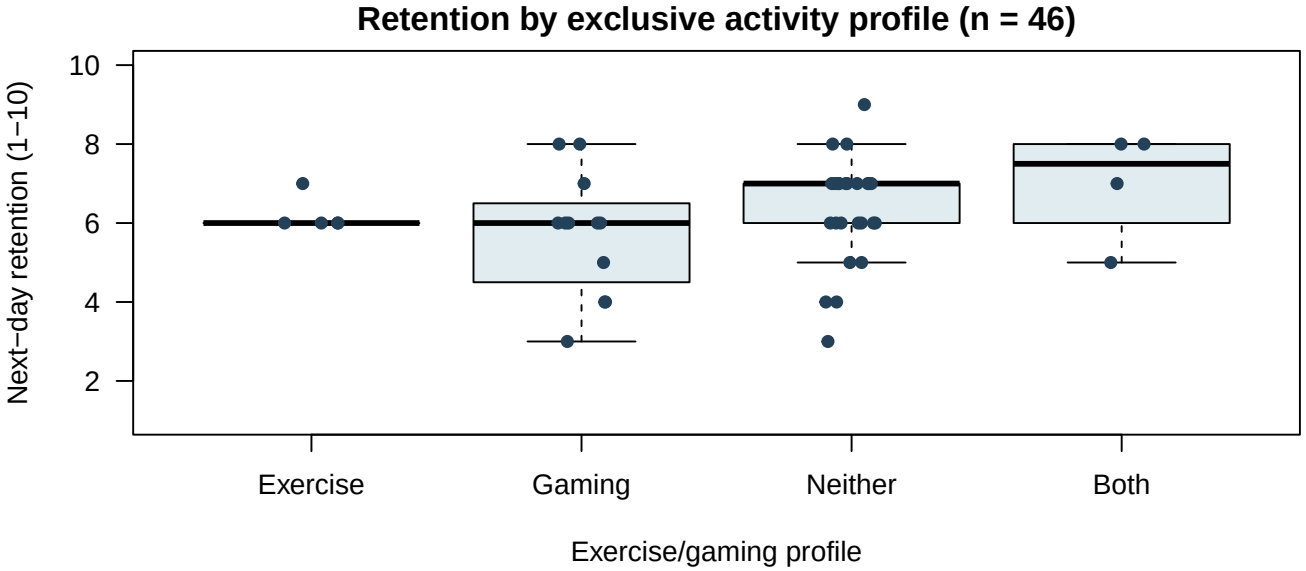


Figure 5: Retention across exclusive activity profiles. Boxes show medians and quartiles; points show individual observations with horizontal jitter only. Exercise and gaming labels indicate the corresponding exclusive profiles. The "both" group has the highest mean but only four respondents, and distributions overlap considerably.

Test $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ against at least one unequal mean. The planned one-way ANOVA gives $F(3, 42) = 0.977$, $p = 0.4128$, with $\eta^2 = 0.065$. It does not reject equality at 0.05. The higher observed mean in the "both" profile is not a statistically established difference. No post-hoc comparisons or additional rank-based or unequal-variance tests are performed. This comparison does not isolate the effect of a single activity.

7 Model assumption and influence checks

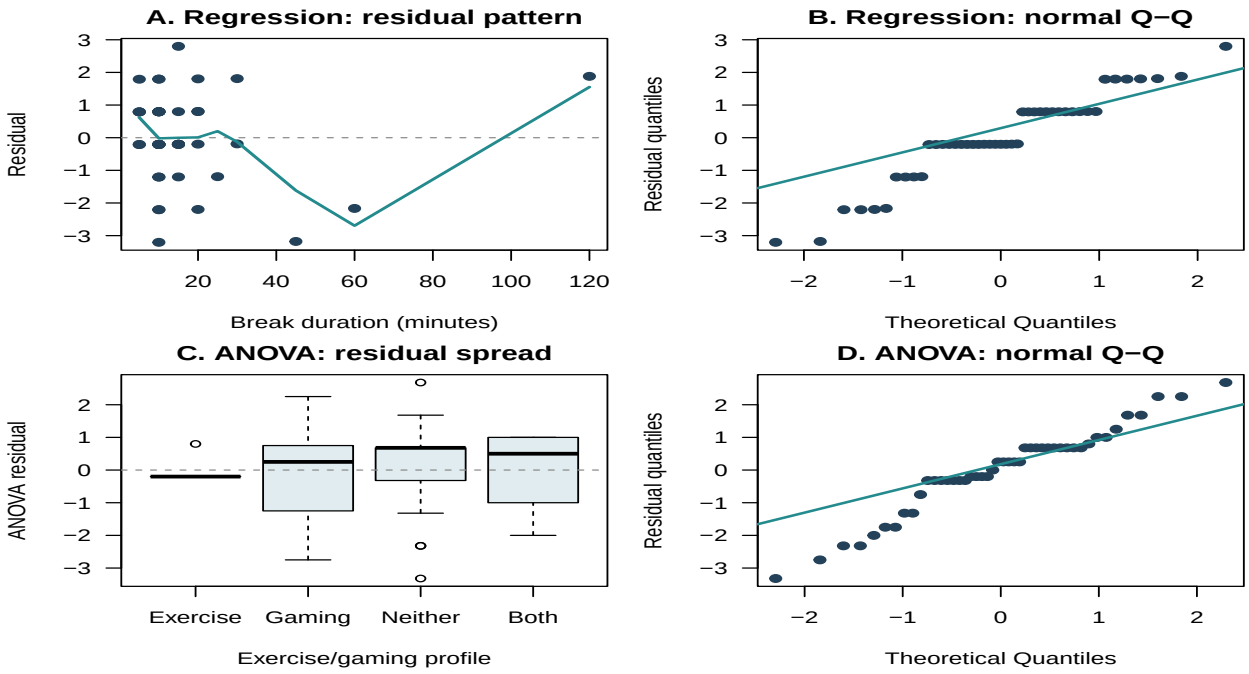


Figure 6: Diagnostics for RQ2 and RQ3. A: RQ2 residual pattern and spread against duration. B: its normal Q–Q plot. C: ANOVA residual spread across profiles. D: ANOVA normal Q–Q plot. Tied ratings and sparse long breaks limit the plots' resolution. RQ1 diagnostics appear in Section 4.

Table 12: Diagnostic tests supplementing the plots.

Model	Check	p
RQ1	Shapiro-Wilk residual normality	0.00085
RQ1	Variance score test	0.00484
RQ2	Shapiro-Wilk residual normality	0.00975
RQ2	Variance score test	0.04998
RQ3	Shapiro-Wilk residual normality	0.06311
RQ3	Median-centered Levene test	0.43691

RQ2 regression. The residual trend is not uniformly flat, Q–Q tails depart from normality, and Shapiro–Wilk rejects ($p = 0.00975$). The variance score test is just below 0.05 ($p = 0.04998$), giving borderline evidence against equal variance. Conventional regression inference is consequently approximate. The retained 120-minute response (record 25) has leverage 0.705 and Cook's distance 7.60, indicating strong influence. It is not deleted to change significance.

ANOVA. Neither normality nor median-centered Levene's test rejects at 0.05. This does not prove the assumptions, especially with groups of four and five. Independence is assumed from individual survey responses and exclusive profile membership; neither diagnostic plots nor ID screening can verify it. The course permits reporting violated assumptions while proceeding with the planned analysis; results are interpreted with those limitations.

8 Interpretation, alignment and limitations

RQ1. The fitted association is positive (0.19377 agreement points per additional break), but not significant (raw $p = 0.0594$; Holm $p = 0.1783$). Residual normality and variance departures further limit the classical inference.

RQ2. The fitted duration slope is near zero and not significant ($p = 0.9445$). Residual departures and one highly influential response limit inference. The result does not prove that duration has no relationship with retention and does not identify an optimal break length.

RQ3. The planned ANOVA does not find a significant mean difference across the four exclusive exercise/gaming profiles ($p = 0.4128$). The largest observed group mean is descriptive and based on only four responses.

Alignment with the revised proposal. RQ1 and RQ2 use simple linear regression; RQ3 uses one-way ANOVA of exclusive exercise/gaming profiles. Raw and Holm-adjusted p-values are reported. All three adjusted results exceed 0.05. Diagnostic checks support the EDA. The proposal's sensitivity analyses remain follow-up work, not completed results in this report.

Limitations.

1. **Sampling and eligibility.** Participation was voluntary. Self-selection and screening may introduce bias, and remaining course IDs are not roster-verified. Results do not represent a validated random sample of the full course.

2. **Preparation.** The missing-ID and timing exclusions, structural missing durations for zero-break cases, and variable-level weekly-hour exclusions are transparent preparation choices. Flags alone do not prove invalidity. Unresolved ranges remain missing. These choices can affect the analysis sample.

3. **Measurement.** Retention is a bounded ordinal self-rating, not an objective test score; agreement is an attitude. Grades are band proxies. Treating the ratings numerically is an approximation.

4. **Year/attention-check item.** Five of 48 respondents selected code 1. If the administered prompt instructed respondents to select "First Year," the other 43 responses did not follow that instruction. This interpretation requires verification against the actual form. These responses are not used as verified years of study or as an additional exclusion rule.

5. **Confounding and precision.** Session length, study load and personal differences are not adjusted for. Small activity groups, co-occurring choices and influential long breaks limit inference. None of the analyses supports a causal learning-improvement claim.

Reproducibility. The accompanying R script generates the summary tables, model results, diagnostic tests, figures and LaTeX value macros. The report is written entirely in `report.tex`; no original report pages are embedded. Appendix A contains the analysis code. Software versions and the datawizard citation are exported to the source folder.

References. Revised group proposal (`proposal_reference.txt`); STA304 Exploratory Data Analysis Instructions (2026). Patil et al. (2022), *datawizard: An R Package for Easy Data Preparation and Statistical Transformations*, JOSS 7(78), 4684, doi:10.21105/joss.04684.

A Complete reproducible R code

Run `Rscript analysis.R` from the source folder after installing `datawizard`, `car` and `jsonlite`. The script reads `survey_source.csv` without modifying it. `verify.R` separately checks key results and that the source data remain unchanged.

```
# STA304 EDA. Run Rscript analysis.R from the submission folder.
# Input: unchanged copy of the new 57-record CSV and its codebook.
# This script computes statistics, preparation and graphics in R.
options(stringsAsFactors = FALSE, width = 78)
needed <- c("datawizard", "car", "jsonlite")
if (!all(vapply(needed, requireNamespace, logical(1), quietly = TRUE))) {
  stop("Install the required packages: ", paste(needed, collapse = ", "))
}
dir.create("tables", showWarnings = FALSE)
dir.create("figures", showWarnings = FALSE)
raw <- read.csv("survey_source.csv", na.strings = "", check.names = FALSE,
  fileEncoding = "UTF-8-BOM", colClasses = c(studentID = "character"))
raw$record <- seq_len(nrow(raw))
raw$bad_id <- is.na(raw$studentID) | !nzchar(trimws(raw$studentID))
total <- with(raw, num_breaks * break_duration_min)
raw$bad_time <- !is.na(total) & total >= raw$study_session_min
raw$bad_week <- !is.na(raw$weekly_study_hours) &
  (raw$weekly_study_hours > 168 | raw$weekly_study_hours < 0)
raw$excluded <- raw$bad_id | raw$bad_time
d <- raw[!raw$excluded, ]
d$weekly_study_hours[d$bad_week] <- NA_real_
# No-break respondents have no defined duration of an actual break.
# Do not infer a duration of zero from the contradictory positive answer.
d$break_duration_min[d$num_breaks == 0] <- NA_real_
split_multi <- function(x) {
  lapply(x, function(z) if (is.na(z)) character() else
    strsplit(z, ",\\s*(?![^()]*\\))", perl = TRUE)[[1]])
}
activities <- split_multi(d$break_activities)
has <- function(label) vapply(activities, function(z) label %in% z, logical(1))
labels <- c(study_session_min = "Study session duration (min)",
  num_breaks = "Breaks per study session",
  break_duration_min = "Duration of each break (min)",
  memory_retention = "Next-day retention rating (1-10)",
  weekly_study_hours = "Weekly study time (hours)",
  prior_grade_pct = "Prior grade: band proxy (%)",
  break_helps_efficiency = "Break benefit agreement (1-7)")
ds <- as.data.frame(datawizard::describe_distribution(d[names(labels)],
  centrality = "all"))
# Fix the quantile convention across package versions.
ds$IQR <- vapply(d[ds$Variable], IQR, numeric(1), na.rm = TRUE, type = 6)
write.csv(ds, "tables/easystats_summary.csv", row.names = FALSE)
num <- data.frame(Variable = unname(labels[ds$Variable]), n = ds$n,
  Missing = ds$n_Missing, Mean = ds$Mean, Median = ds$Median,
  SD = ds$SD, IQR = ds$IQR,
  Min = vapply(d[ds$Variable], min, numeric(1), na.rm = TRUE),
  Max = vapply(d[ds$Variable], max, numeric(1), na.rm = TRUE))
freq <- function(x, levels, labels = levels) {
  z <- table(factor(x, levels = levels), useNA = "always")
  data.frame(Category = c(labels, "Missing / unavailable"),
    n = as.integer(z), Percent = 100 * as.integer(z) / length(x))
}
year <- freq(d$year_of_study, 1:5, paste("Recorded code", 1:5))
grade <- freq(d$prior_grade_pct, c(55, 65, 75, 85, 95),
  c("50-59%", "60-69%", "70-79%", "80-89%", "90-100%"))
```

```
agree <- freq(d$break_helps_efficiency, 1:7,
  c("Strongly disagree", "Disagree", "Slightly disagree", "Neutral",
    "Slightly agree", "Agree", "Strongly agree"))
memory <- freq(d$memory_retention, 1:10, as.character(1:10))
al <- sort(unique(unlist(activities)))
act <- data.frame(Category = al,
  n = vapply(al, function(z) sum(has(z)), integer(1)))
act$Category[act$Category == "Rotting on instagram specifically"] <-
  "Instagram write-in"
act$Percent <- 100 * act$n / nrow(d)
loc <- split_multi(d$study_location)
loc <- lapply(loc, function(z) {
  z[z %in% c("UTM buildings like IB or MN",
    "University Buildings (e.g. MN)")] <- "Other campus building"
  z[z == "Lecture hall(but self-study in lecture hall)"] <- "Lecture hall"
  z[z == "Transit(in bus, subway, train, etc)"] <- "Public transit"
  z[z == "Outdoors(park, beach, etc)"] <- "Outdoors"
  unique(z)
})
ll <- sort(unique(unlist(loc)))
location <- data.frame(Category = c(ll, "Missing / unusable"),
  n = c(vapply(ll, function(z) sum(vapply(loc,
    function(v) z %in% v, logical(1))), integer(1)),
    sum(lengths(loc) == 0)))
location$Percent <- 100 * location$n / nrow(d)
ex <- has("Exercise"); ga <- has("Gaming")
d$activity_profile <- factor(ifelse(ex & ga, "Both",
  ifelse(ex, "Exercise only", ifelse(ga, "Gaming only", "Neither"))),
  levels = c("Exercise only", "Gaming only", "Neither", "Both"))
breakers <- droplevels(d[d$num_breaks > 0, ])
bt <- breakers[complete.cases(breakers[c("break_duration_min",
  "memory_retention")]), ]
ad <- droplevels(breakers[!is.na(breakers$memory_retention), ])
# RQ1 follows the revised proposal: retain the numerical count and rating.
rq1 <- lm(break_helps_efficiency ~ num_breaks, data = d)
rq1_sm <- summary(rq1)
rq1_ci <- confint(rq1)
rq1_shapiro <- shapiro.test(residuals(rq1))
rq1_ncv <- car::ncvTest(rq1)
ols <- lm(memory_retention ~ break_duration_min, data = bt)
av <- aov(memory_retention ~ activity_profile, data = ad)
sm <- summary(ols); avtab <- summary(av)[[1]]; ci <- confint(ols)
sh_ols <- shapiro.test(residuals(ols)); ncv <- car::ncvTest(ols)
sh_av <- shapiro.test(residuals(av))
lev <- car::leveneTest(memory_retention ~ activity_profile,
  data = ad, center = median)
coeff <- data.frame(Term = c("Intercept", "Break duration (min)"),
  Estimate = unname(coef(ols)), SE = sm$coefficients[, 2],
  t = sm$coefficients[, 3], p = sm$coefficients[, 4],
  Lower = ci[, 1], Upper = ci[, 2], row.names = NULL)
fit <- data.frame(n = nrow(bt), Residual_df = df.residual(ols),
  R_squared = sm$r.squared, Adjusted_R_squared = sm$adj.r.squared,
  Residual_SE = sm$sigma, F = unname(sm$fstatistic[1]))
tests <- data.frame(Question = c("RQ1", "RQ2", "RQ3"),
  Test = c("Simple linear regression", "Simple linear regression", "One-way ANOVA"),
  n = c(nrow(d), nrow(bt), nrow(ad)),
  Statistic = c(rq1_sm$coefficients[2, 3], sm$coefficients[2, 3],
    avtab[1, "F value"]),
  p = c(rq1_sm$coefficients[2, 4], sm$coefficients[2, 4], avtab[1, "Pr(>F)"]))
tests$Holm_p <- p.adjust(tests$p, method = "holm")
groups <- do.call(rbind, lapply(split(ad$memory_retention,
```

```
  ad$activity_profile), function(z) c(n = length(z), Mean = mean(z),
    SD = sd(z), Median = median(z))))
groups <- data.frame(Profile = rownames(groups), groups, row.names = NULL)
checks <- data.frame(Model = c("RQ1", "RQ1", "RQ2", "RQ2", "RQ3", "RQ3"),
  Check = rep(c("Shapiro-Wilk residual normality", "Variance score test"), 3),
  p = c(rq1_shapiro$p.value, rq1_ncv$p, sh_ols$p.value, ncv$p,
    sh_av$p.value, lev[1, "Pr(>F)"]))
checks$Check[6] <- "Median-centered Levene test"
rq1_coeff <- data.frame(Term = c("Intercept", "Break count"),
  Estimate = unname(coef(rq1)), SE = rq1_sm$coefficients[, 2],
  t = rq1_sm$coefficients[, 3], p = rq1_sm$coefficients[, 4],
  Lower = rq1_ci[, 1], Upper = rq1_ci[, 2], row.names = NULL)
cooks <- cooks.distance(ols)
influence <- data.frame(record = bt$record, fitted = fitted(ols),
  residual = residuals(ols), standardized_residual = rstandard(ols),
  leverage = hatvalues(ols), cooks_distance = cooks)
safe <- d[setdiff(names(d), c("studentID", "timestamp", "data_notes"))]
write.csv(safe, "analysis_ready.csv", row.names = FALSE, na = "")
write.csv(bt[c("record", "break_duration_min", "memory_retention")],
  "regression_data.csv", row.names = FALSE)
write.csv(raw[c("record", "bad_id", "bad_time", "bad_week", "excluded")],
  "tables/record_audit.csv", row.names = FALSE)
write.csv(influence, "tables/regression_influence.csv", row.names = FALSE)
write.csv(fit, "tables/model_fit.csv", row.names = FALSE)
all_tables <- list(rq1_coefficients = rq1_coeff, numeric = num, year = year, grade = grade,
  agreement = agree, retention = memory, activities = act,
  locations = location, model_coefficients = coeff, primary_tests = tests,
  group_summary = groups, assumption_checks = checks)
tex_escape <- function(x) {
  x <- gsub("%", "\\%", x, fixed = TRUE)
  x <- gsub("&", "\\&", x, fixed = TRUE)
  gsub("_", "\\_", x, fixed = TRUE)
}
for (nm in names(all_tables)) {
  tb <- all_tables[[nm]]
  write.csv(tb, paste0("tables/", nm, ".csv"), row.names = FALSE)
  display <- tb
  for (cn in names(tb)) {
    if (is.numeric(tb[[cn]])) {
      dp <- if (cn %in% c("n", "Missing")) 0 else
        if (cn == "Percent") 1 else
        if (nm %in% c("model_coefficients", "rq1_coefficients", "assumption_checks")) 5 else
        if (cn %in% c("p", "Holm_p")) 4 else 2
      display[[cn]] <- formatC(tb[[cn]], format = "f", digits = dp)
      if (cn == "p") display[[cn]][tb[[cn]] < .00001] <- "$<0.00001$"
    }
    display[[cn]] <- tex_escape(display[[cn]])
  }
  lines <- apply(display, 1, function(z)
    paste0(paste(z, collapse = " & "), " \\\\"))
  writeLines(c(lines, "\\bottomrule"), paste0("tables/", nm, ".tex"))
}
values <- c(Intercept = sprintf("%.5f", coef(ols)[1]),
  Slope = sprintf("%.6f", coef(ols)[2]),
  SlopeP = sprintf("%.4f", coeff$p[2]),
  SlopeT = sprintf("%.3f", coeff$t[2]),
  SlopeLo = sprintf("%.5f", ci[2, 1]),
  SlopeHi = sprintf("%.5f", ci[2, 2]),
  RSquared = sprintf("%.6f", sm$r.squared),
  AdjRSquared = sprintf("%.5f", sm$adj.r.squared),
  ResidualSE = sprintf("%.3f", sm$sigma),
```

```
  FStat = sprintf("%.4f", unname(sm$fstatistic[1])),
  TenEffect = sprintf("%.4f", 10 * coef(ols)[2]),
  TenLo = sprintf("%.4f", 10 * ci[2, 1]),
  TenHi = sprintf("%.4f", 10 * ci[2, 2]),
  ShapiroP = sprintf("%.5f", sh_ols$p.value),
  NCVP = sprintf("%.5f", ncv$p),
  MaxCook = sprintf("%.2f", max(cooks)),
  MaxHat = sprintf("%.3f", max(hatvalues(ols))),
  RQOneSlope = sprintf("%.5f", coef(rq1)[2]),
  RQOneP = sprintf("%.4f", rq1_sm$coefficients[2, 4]),
  RQOneLo = sprintf("%.5f", rq1_ci[2, 1]),
  RQOneHi = sprintf("%.5f", rq1_ci[2, 2]),
  RQOneRSquared = sprintf("%.4f", rq1_sm$r.squared),
  RQOneShapiro = sprintf("%.5f", rq1_shapiro$p.value),
  RQOneNCV = sprintf("%.5f", rq1_ncv$p),
  RQOneHolm = sprintf("%.4f", tests$Holm_p[1]),
  AnovaF = sprintf("%.3f", avtab[1, "F value"]),
  AnovaP = sprintf("%.4f", avtab[1, "Pr(>F)"]),
  EtaSquared = sprintf("%.3f", avtab[1, "Sum Sq"] / sum(avtab[, "Sum Sq"])))
names(values) <- sub("\\..*$", "", names(values))
writeLines(paste0("\\newcommand{\\", names(values), "}{", values, "}"),
  "tables/report_values.tex")
memory_lines <- c(
  paste0("Count & ", paste(memory$n[1:10], collapse = " & "), " \\\\"),
  paste0("\\% of 48 & ", paste(sprintf("%.1f", memory$Percent[1:10]),
    collapse = " & "), " \\\\"), "\\bottomrule")
writeLines(memory_lines, "tables/retention_wide.tex")
navy <- "#23425A"; teal <- "#238A8D"; red <- "#B44B45"
pdf("figures/rq1_results.pdf", width = 7.2, height = 2.5)
par(mar = c(4, 4, 1, 1), cex = .85, las = 1)
set.seed(304)
plot(jitter(d$num_breaks, amount = .07),
  jitter(d$break_helps_efficiency, amount = .05), pch = 19, col = navy,
  xlab = "Breaks per study session", ylab = "Benefit agreement (1-7)")
abline(rq1, col = teal, lwd = 2)
dev.off()
pdf("figures/rq1_diagnostics.pdf", width = 7.2, height = 2.5)
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1), cex = .8, las = 1)
plot(fitted(rq1), residuals(rq1), pch = 19, col = navy,
  xlab = "Fitted agreement", ylab = "Residual", main = "RQ1: residual spread")
abline(h = 0, lty = 2, col = "grey55")
lines(lowess(fitted(rq1), residuals(rq1)), col = teal, lwd = 2)
qqnorm(residuals(rq1), pch = 19, col = navy, main = "RQ1: normal Q-Q")
qqline(residuals(rq1), col = teal, lwd = 2)
dev.off()
pdf("figures/descriptive_distributions.pdf", width = 7.2, height = 2.65)
par(mfrow = c(1, 2), mar = c(4.1, 3.9, 2.2, .7), cex = .82, las = 1)
hist(bt$break_duration_min, breaks = seq(0, 125, 5), col = "#DCE9EE",
  border = "white", main = "A. Break duration (n = 45)",
  xlab = "Duration of each break (minutes)", ylab = "Respondents")
barplot(as.integer(table(factor(d$memory_retention, levels = 1:10))),
  names.arg = 1:10, col = "#DCE9EE", border = NA,
  main = "B. Retention rating (n = 48)",
  xlab = "Next-day retention rating (1-10)", ylab = "Respondents")
dev.off()
pdf("figures/research_results.pdf", width = 7.2, height = 3.65)
par(mar = c(4.2, 4.2, 2.1, .8), cex = .92, las = 1)
grid <- data.frame(break_duration_min = seq(min(bt$break_duration_min),
  max(bt$break_duration_min), length.out = 250))
prediction <- predict(ols, grid, interval = "confidence")
plot(NA, xlim = c(0, 128), ylim = c(1, 10),
```

```
  xlab = "Duration of each break (minutes)",
  ylab = "Next-day retention rating (1-10)",
  main = "Break duration and perceived next-day retention (n = 45)")
polygon(c(grid$break_duration_min, rev(grid$break_duration_min)),
  c(prediction[, "lwr"], rev(prediction[, "upr"])),
  col = adjustcolor(teal, alpha.f = .15), border = NA)
lines(grid$break_duration_min, prediction[, "fit"], col = teal, lwd = 2)
points_n <- aggregate(rep(1, nrow(bt)),
  list(Minutes = bt$break_duration_min, Rating = bt$memory_retention), sum)
points(points_n$Minutes, points_n$Rating, pch = 21,
  bg = adjustcolor(navy, alpha.f = .7), col = navy,
  cex = .85 * sqrt(points_n$x))
im <- which.max(cooks)
points(bt$break_duration_min[im], bt$memory_retention[im],
  pch = 21, col = red, bg = red, cex = 1.1)
text(bt$break_duration_min[im], bt$memory_retention[im],
  labels = "Record 25", pos = 2, col = red, cex = .83)
legend("bottomright", c("1 respondent", "3 respondents", "6 respondents"),
  pch = 21, pt.bg = navy, col = navy, pt.cex = .85 * sqrt(c(1, 3, 6)),
  bty = "n", cex = .78, y.intersp = 1.2)
dev.off()
short <- c("Exercise", "Gaming", "Neither", "Both")
pdf("figures/activity_comparison.pdf", width = 7.2, height = 3.2)
par(mar = c(4.2, 4, 2, .8), cex = .9, las = 1)
boxplot(memory_retention ~ activity_profile, data = ad, names = short,
  col = "#E0ECEF", xlab = "Exercise/gaming profile",
  ylab = "Next-day retention (1-10)", ylim = c(1, 10),
  main = "Retention by exclusive activity profile (n = 46)", outline = FALSE)
set.seed(304)
stripchart(memory_retention ~ activity_profile, data = ad,
  method = "jitter", vertical = TRUE, add = TRUE, pch = 19, col = navy)
dev.off()
pdf("figures/model_diagnostics.pdf", width = 7.2, height = 5.6)
par(mfrow = c(2, 2), mar = c(4, 4, 2.1, 1), cex = .85, las = 1)
plot(bt$break_duration_min, residuals(ols), pch = 19, col = navy,
  xlab = "Break duration (minutes)", ylab = "Residual",
  main = "A. Regression: residual pattern")
abline(h = 0, col = "grey55", lty = 2)
lines(lowess(bt$break_duration_min, residuals(ols)), col = teal, lwd = 2)
qqnorm(residuals(ols), pch = 19, col = navy,
  main = "B. Regression: normal Q-Q", ylab = "Residual quantiles")
qqline(residuals(ols), col = teal, lwd = 2)
boxplot(residuals(av) ~ ad$activity_profile, names = short, col = "#E0ECEF",
  xlab = "Exercise/gaming profile", ylab = "ANOVA residual",
  main = "C. ANOVA: residual spread")
abline(h = 0, col = "grey55", lty = 2)
qqnorm(residuals(av), pch = 19, col = navy,
  main = "D. ANOVA: normal Q-Q", ylab = "Residual quantiles")
qqline(residuals(av), col = teal, lwd = 2)
dev.off()
pdf("figures/data_quality.pdf", width = 7.2, height = 3.35)
par(mfrow = c(1, 2), mar = c(4, 4, 2.2, 1), cex = .84, las = 1)
ok <- !is.na(total) & total > 0
plot(raw$study_session_min[ok], total[ok], log = "xy", pch = 19,
  col = ifelse(raw$bad_time[ok], red, navy),
  xlab = "Session minutes (log scale)",
  ylab = "Total break minutes (log scale)", main = "A. Session-time consistency")
abline(0, 1, col = "grey55", lty = 2)
extreme <- which(raw$bad_time & total > 1000)
text(raw$study_session_min[extreme], total[extreme],
  labels = paste("Record", extreme), pos = 2, cex = .8)
```

```
wk <- !is.na(raw$weekly_study_hours) & raw$weekly_study_hours > 0
plot(raw$record[wk], raw$weekly_study_hours[wk], log = "y", pch = 19,
  col = ifelse(raw$bad_week[wk], red, navy), xlab = "Source record number",
  ylab = "Weekly study time (hours; log scale)", main = "B. Weekly-time outliers")
abline(h = 168, lty = 2, col = red)
text(18, 168, "168 hours/week", pos = 3, cex = .8, col = red)
dev.off()
result <- list(method_reference = "proposal_reference.txt",
  raw_n = nrow(raw), screened_n = nrow(d),
  break_takers_n = nrow(breakers), regression_n = nrow(bt), anova_n = nrow(ad),
  bad_id_records = raw$record[raw$bad_id],
  timing_records = raw$record[raw$bad_time],
  weekly_outlier_records = raw$record[raw$bad_week],
  numeric_summary = num, coefficients = coeff, model_fit = fit,
  primary_tests = tests, activity_groups = groups, assumptions = checks,
  rq1 = list(coefficients = rq1_coeff, r_squared = rq1_sm$r.squared,
    residual_df = df.residual(rq1), shapiro_p = rq1_shapiro$p.value,
    variance_score_p = rq1_ncv$p),
  anova = list(df1 = avtab[1, "Df"], df2 = avtab[2, "Df"],
    eta_squared = avtab[1, "Sum Sq"] / sum(avtab[, "Sum Sq"])),
  influence = list(max_cook_record = bt$record[im],
    max_cook = max(cooks), max_leverage = max(hatvalues(ols))))
jsonlite::write_json(result, "results.json", auto_unbox = TRUE,
  pretty = TRUE, digits = 15)
capture.output(sessionInfo(), file = "session_info.txt")
capture.output(citation("datawizard"), file = "tables/datawizard_citation.txt")
```

B Source-data consistency and outlier plots

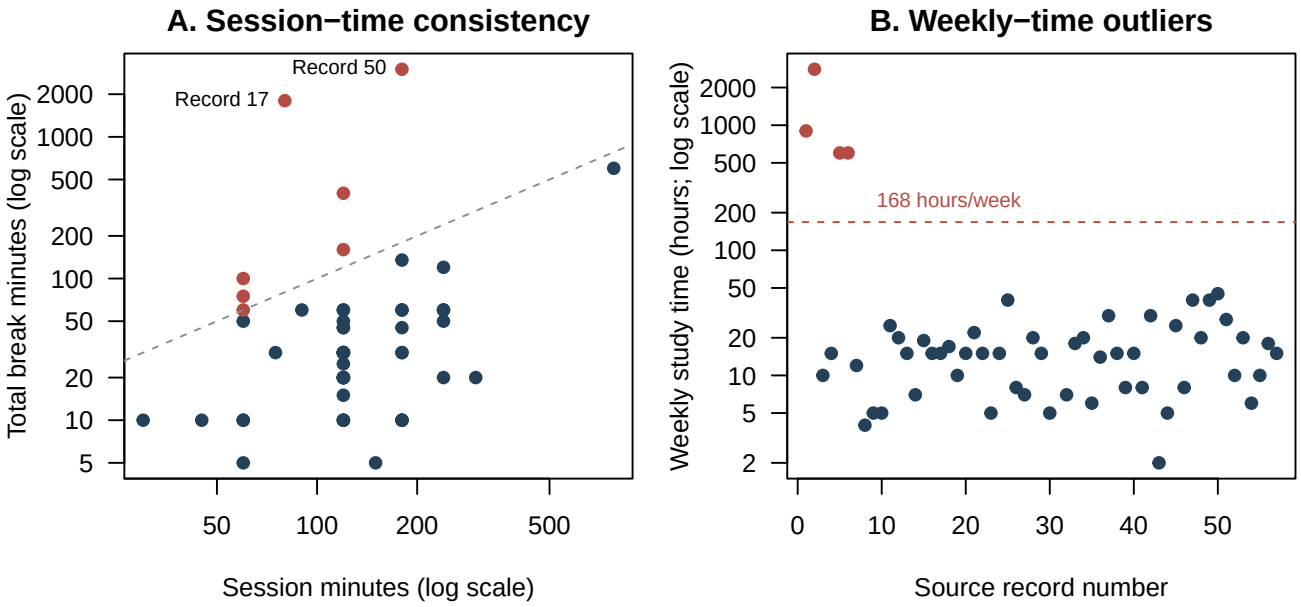


Figure 7: Source-data consistency checks. A: the 54 positive, observed total break times against session duration on log scales. The seven red points meet or exceed the equality line and are excluded by the timing screen. Two no-break zero totals and one missing duration are not shown on the log plot. Only the two largest inconsistent totals are labelled. B: the 56 observed weekly-hour entries, including four above the 168-hour limit. Those four fields are set to missing, without assuming a unit conversion. Source-record numbers start from 1 after the CSV header.